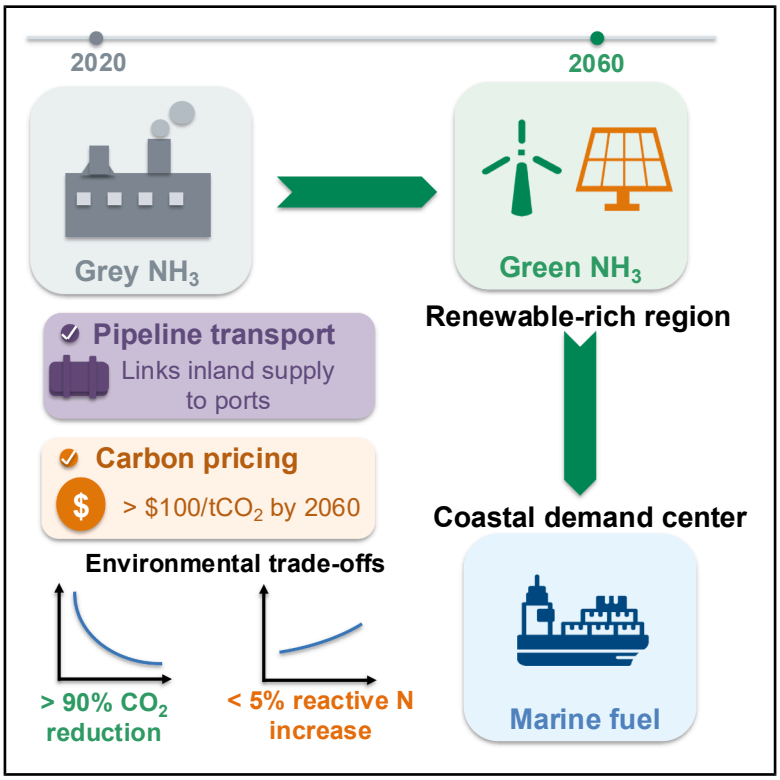
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Article

Demand for clean marine fuel reshapes China's ammonia supply chains with viability hinging on pipelines and carbon prices

Graphical abstract



Highlights

- Spatial mismatch exists between green ammonia supply and emerging demand in China
- Pipeline transportation reduces ammonia costs by ~18% in 2060 versus trucks
- Ammonia fuel needs pipelines and >$100/$tCO_2$ carbon price to be economically viable
- Reactive-nitrogen emissions from ammonia can be managed below 5% of current totals

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In brief

Transitioning to green ammonia is becoming more common as marine fuel faces significant cost hurdles and geographic mismatches in China. This research analyzes how demand reshapes spatiotemporal supply chains. It underscores that pipelines and carbon pricing are essential for viability, providing a scalable roadmap to decarbonize both ammonia production and shipping sectors.

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Demand for clean marine fuel reshapes China's ammonia supply chains with viability hinging on pipelines and carbon prices

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SCIENCE FOR SOCIETY Ammonia is well known as a key ingredient in fertilizer, but it is also emerging as a competitive low-carbon fuel for shipping. Nevertheless, most ammonia today is produced from coal and natural gas, offsetting its potential climate benefits. Shifting to green ammonia, which is synthesized using renewable energy, is essential, but it faces significant hurdles such as high costs for emerging technologies and the geographic mismatch between renewable resources and future fuel demand. A key but unanswered question is how will growing demand for clean marine fuel influence the green ammonia transition? Using China as a case study, our analysis shows that, to meet clean marine fuel demand, large-scale deployment of green ammonia could take place as early as around 2030. Achieving cost-competitive green ammonia for marine fuel will require pipeline transportation for improved supply-demand matching and carbon prices to rise above \$100/tCO_2. Coordinated actions taken by policymakers and stakeholders on technology support, facility siting, and carbon pricing will be critical for a smooth and effective transition to green ammonia.

SUMMARY

Green ammonia is taking center stage as a low-carbon marine fuel, driving the shipping sector's net-zero transition. However, the ramifications of soaring demand for clean marine fuel on today's carbon-intensive ammonia production and the spatiotemporal evolution of ammonia supply chains are far from clear. Here, we develop a technology-rich, multi-region model for China optimizing production routes, facility locations, and transportation under various policy and infrastructure scenarios. The results reveal that large-scale deployment of green hydrogen to ammonia could begin around 2030, cutting CO_2 emissions by >90% by 2050, though supply costs will rise 6%–30% above 2020 levels. Pipelines could cut costs by 18% compared with trucks by improving supply-demand matching, and carbon prices above 100 USD/t CO_2 are needed to make green ammonia viable as a marine fuel by 2060. Our findings offer a feasible green ammonia road map for China with strategic priorities for both policy and infrastructure development.

INTRODUCTION

Mitigating climate change will necessitate a profound transformation in ammonia production and use. Ammonia is primarily synthesized from fossil energy and underpins fertilizer supply and a wide range of chemical products.[1] Globally, ammonia production is responsible for about 2% of greenhouse gas (GHG) emissions, releasing around 420 million metric tons of carbon dioxide (Mt CO_2) in 2022,[2] a level comparable to the total GHG emissions of the United Kingdom.[3] Decarbonizing ammonia production is therefore critical for meeting climate targets while sustaining agricultural productivity and related chemical value chains.[4] At the same time, low-carbon ammonia is increasingly discussed as a candidate zero-carbon fuel to help meet the

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International Maritime Organization's (IMO's) target of net-zero GHG emissions from international shipping by or around 2050.[5,6] Together, these developments position ammonia as a nexus linking energy, agriculture, industry, and shipping systems, and its decarbonization requires a coherent supply-chain perspective.

The transition of ammonia faces economic and environmental challenges on both the supply and the demand sides. On the supply side, process emissions associated with carbonaceous feedstocks are difficult to eliminate for incumbent fossil-based routes.[7] Hence, achieving deep reductions ultimately requires shifting to nascent low-carbon technologies such as green-hydrogen-based routes and fossil-based routes with carbon capture and storage (CCS), which currently carry substantial cost premiums.[8] On the demand side, ammonia fuel can deliver meaningful well-to-wake emissions reductions only if its production is decarbonized.[9,10] Even fossil-based ammonia is generally more expensive than oil-based marine fuels, and the additional cost premiums of low-carbon ammonia further exacerbate the economic challenges of large-scale fuel switching in shipping.[11] The viability of ammonia fuel also hinges on optimally balancing production and logistics, including the choice of transport mode,[12] given the potential spatial mismatch between low-cost production regions and coastal demand. Meanwhile, using ammonia fuel introduces environmental risks, particularly via reactive-nitrogen emissions (NH_3, N_2O, and NO_x).[13,14] Technology shifts, demand restructuring, and interregional transport and infrastructure constraints should therefore be integrated within a supply-chain framework to evaluate technology adoption, end-use affordability, and environmental benefits and risks.

Existing studies provide limited insight into how emerging marine-fuel demand reshapes ammonia supply chains across space and time. Some quantify the levelized cost of ammonia (LCOA), focusing either on production-route benchmarking[15,16] or on grid-interactive assessments,[17–19] and highlight how resource endowments drive production costs but stop short of analyzing delivered costs. Others optimize the spatial topology of ammonia transport networks and supply chains at regional or global scales[20–22] yet often focus only on green ammonia and do not fully capture long-term capacity evolution and competition among production routes. A third stream explores single-zone[23–25] or multi-zone[26,27] decarbonization pathways but does not explicitly incorporate ammonia fuel demand, leaving unclear how this emerging demand feeds back on spatiotemporal technology deployment, supply-chain configuration, and transport infrastructure investment. Consequently, the spatiotemporal evolution of delivered ammonia costs and environmental impacts along realistic supply chains remains poorly understood. A key knowledge gap concerns the technoeconomic, policy, and infrastructure conditions under which ammonia can serve as an economically viable and environmentally robust marine fuel option.

China provides an ideal case to study how ammonia supply chains might be reshaped under simultaneous production decarbonization and fuel transition pressures. The carbon neutrality pledge by 2060 in China is catalyzing broader interest in low-carbon ammonia production and consumption, including demonstration projects for green ammonia supplying shipping fuel.[28] As the world's largest ammonia producer and consumer, China accounts for roughly one-third of global output, with a technology mix still dominated by coal-based routes[29,30] (see Note S1 for a detailed overview of China's ammonia industry). China also hosts some of the world's largest port clusters and shipping lanes,[31] positioning it as a central player in the prospective deployment of low-carbon marine fuels (see Note S2 for China's shipping sector). At the same time, China exhibits pronounced spatial mismatch between low-carbon resource endowments and ammonia demand centers. High-quality wind and solar resources and major geological CO_2 storage basins are concentrated in northern inland regions,[32,33] whereas maritime activity is concentrated in eastern and southeastern coastal provinces. Similar spatial mismatches are also observed in other large economies[34,35] (see Note S3 for details).

In this study, we explore the spatiotemporal evolution of the supply chain and industrial layout of China's ammonia sector and evaluate the economic implications and environmental impacts of alternative decarbonization pathways (see Table 1 and Note S4), with particular emphasis on ammonia as a marine fuel. We develop a multi-region, multi-stage planning model that integrates a bottom-up ammonia demand forecast with an optimization-based, technology-rich supply module, using an hourly resolution technoeconomic database as input. We find that regional supply-demand imbalances of ammonia are likely to be exacerbated in China by 2060, requiring transportation from production centers (primarily green-hydrogen-based technologies) in northern regions to demand hubs in southeastern coastal areas. Green-hydrogen-based technologies begin to deploy at scale around 2030, driving nearly 90% of reductions in CO_2 emissions over 2030–2050 but increasing ammonia supply costs by approximately 6%–30% compared with those in 2020 under moderate carbon prices (111 USD/t CO_2 by 2060). Building long-distance ammonia pipelines could reduce delivered costs by around 18% by 2060 and cumulative CO_2 emissions by 3%–13% relative to truck transport, enabling ammonia fuel to become economically viable under moderate carbon prices. With appropriate regulation, the added reactive nitrogen emissions from ammonia fuel can be kept to less than 5% of current national totals. Taken together, our analysis provides a supply-chain-wide assessment of how concurrent transformations on the supply and demand sides reshape China's ammonia decarbonization pathways and offers quantitative evidence on the economic viability and environmental consequences of using ammonia as a marine fuel.

RESULTS

Marine fuel demand widens the coastal-inland mismatch

The consumption of ammonia in China is projected to increase gradually, driven by the rising demand for ammonia as a shipping fuel. Figure 1A shows that, from 2020 to 2060, agricultural ammonia consumption is expected to decline by 25.4%, primarily due to reduced demand for fertilizers. Conversely, industrial consumption is expected to rise to 24.9 Mt by 2060, driven by increased downstream chemical production. The decrease in agricultural demand and the increase in industrial demand are likely to approximately counterbalance each other, resulting in a relatively stable total consumption for conventional ammonia

Table 1. Scenario design and description

Scenario	Fuel transportation method	Renewable availability	Green hydrogen cost	CCS cost	CCS availability	Fuel demand	Carbon tax
TT	tanker truck	medium	medium	medium	yes	yes	medium
PL	pipeline	medium	medium	medium	yes	yes	medium
TT_RE-	tanker truck	low	medium	medium	yes	yes	medium
PL_RE-	pipeline	low	medium	medium	yes	yes	medium
TT_H2+	tanker truck	medium	high	medium	yes	yes	medium
PL_H2+	pipeline	medium	high	medium	yes	yes	medium
TT_CCS+	tanker truck	medium	medium	high	yes	yes	medium
PL_CCS+	pipeline	medium	medium	high	yes	yes	medium
TT_NoCCS	tanker truck	medium	medium	N/A	no	yes	medium
PL_NoCCS	pipeline	medium	medium	N/A	no	yes	medium
TT_NoFD	tanker truck	medium	medium	medium	yes	no	medium
PL_NoFD	pipeline	medium	medium	medium	yes	no	medium
TT_CP-	tanker truck	medium	medium	medium	yes	yes	low
PL_CP-	pipeline	medium	medium	medium	yes	yes	low
TT_CP+	tanker truck	medium	medium	medium	yes	yes	exponential
PL_CP+	pipeline	medium	medium	medium	yes	yes	exponential
Best	pipeline	high	low	low	yes	yes	medium
Worst	tanker truck	low	high	high	no	yes	medium

applications. Domestic demand for ammonia as a shipping fuel is expected to begin around 2026, growing to reach 36.3 Mt by 2060, accounting for ~40% of the total consumption.

The rising demand for shipping fuel prompts a shift in ammonia demand toward the southeastern coastal regions, which are home to large port facilities. Figures 1B–1D depict total provincial demand for ammonia and sectoral composition of demand in 2020, 2040, and 2060. The emerging fuel demand is concentrated in the southeastern coastal provinces and becomes particularly notable in these areas. The total ammonia demand in these coastal regions is expected to reach 53.1 Mt, which is ~2.7 times the demand in 2020. Overall, the southern coastal provinces exhibit higher ammonia fuel demand than the northern coastal provinces due to greater freight turnover. In most coastal provinces, ammonia fuel demand accounts for over 70% of the total demand. For instance, although Shanghai had a relatively low ammonia demand in 2020, as the world's largest port, Shanghai may become one of the largest demand centers for ammonia in China by 2060, requiring 12.1 Mt of ammonia fuel (representing 99.6% of Shanghai's total demand).

China's ammonia production could undergo significant technological transformation and rapid CO_2 emission reduction from 2030 to 2050, as shown in Figures 2B and 2E. Until 2030, coal-based and gas-based routes are still prevalent. Given expected cost reductions for electrolysis and renewable electricity generation technologies, hydrogen-based ammonia production begins to deploy in 2032 and becomes the dominant production method by 2060. CCS technology may be implemented for coal-based routes beginning around 2038, nearly eliminating conventional (non-CCS) coal-based production by 2050. Utilizing pipelines instead of tanker trucks (TTs) for ammonia transportation could increase the use of hydrogen-based production. By 2060, ammonia output from hydrogen-based plants could account for ~80% in the pipelines (PL) and ~67% in the TT scenario. Residual emissions in 2060 are projected to be about 12% of 2021 levels under the TT and 7% under the PL scenario.

The spatial layout of China's ammonia industry will evolve, driven by technological transformation and significantly influenced by the transportation modes of ammonia fuel, as shown in Figures 2A, 2C, 2D, 2F, and 2G. In 2020, ammonia production was concentrated in northern and central provinces rich in fossil energy resources and downstream demand. By 2060, northern regions are expected to become centers of large-scale hydrogen-based ammonia production. Due to its excellent renewable resources, Inner Mongolia is anticipated to become the largest-producing province, with an output of approximately 42 Mt in both TT and PL scenarios. Some southern provinces with viable CO_2 sequestration options could adopt coal-CCS technology on a considerable scale. If TTs are used for fuel transportation, a southern production hub dominated by coal-CCS routes might emerge. For instance, Hunan could produce 9.9 Mt of ammonia under the TT scenario (94% from coal with CCS) but only 1.0 Mt under the PL scenario. This highlights the importance of strengthening pilot projects for nascent technologies tailored to regional resource endowment and technical suitability and considering the impact of transportation modes on the regional evolution of hydrogen production.

A general pattern of urea transportation from the northwest to the southeast will likely intensify as China's ammonia sector decarbonizes. In 2021, urea predominantly moved from Inner Mongolia and Shanxi to northeastern and southeastern provinces, as shown in Figure 3A. The interregional trade of urea under TT and PL exhibits similar characteristics in 2060, illustrated in Figures 3B and 3C. Inner Mongolia and Qinghai might become key exporters of urea. The total equivalent ammonia transport reaches 26.9–28.9 Mt by 2060, which is 2.0–2.2 times that of 2021. This indicates that future urea trade will heavily depend on green urea production from renewable-rich regions.

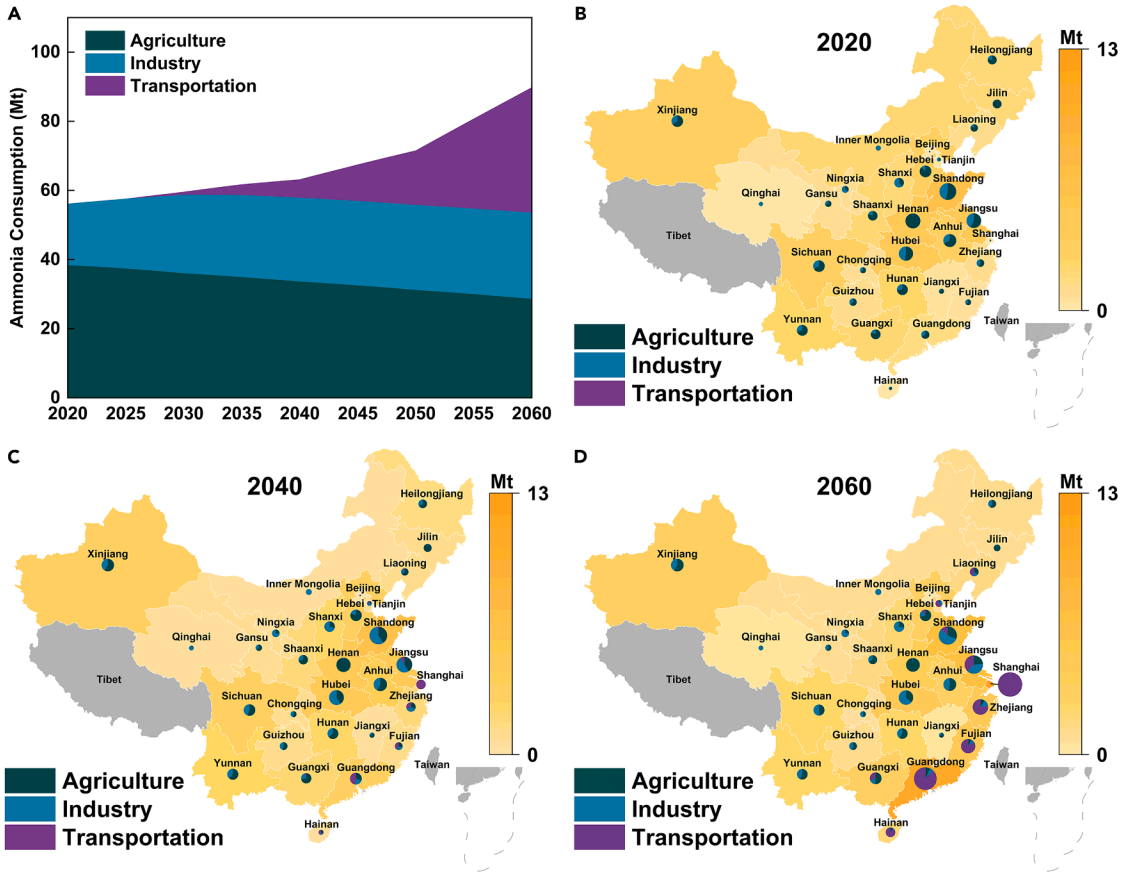


Figure 1. China's ammonia demand structure evolution and spatial distribution

(A) National ammonia consumption structure pathway.

(B–D) Provincial ammonia consumption and structure in 2020, 2040, and 2060. Ammonia consumption refers to the equivalent amount of ammonia for all commodities consumed in a specific region. The base color of the map and the size of the pie charts for each province represent the equivalent ammonia consumption for that year. The pie charts illustrate the structure of each province's consumption that year. Only China's domestic shipping fuel demand is included.

The transport distance and pattern of ammonia fuel can vary significantly under different transportation modes, as shown in Figures 3D and 3E. Local renewable resources are insufficient for ammonia production in most provinces, necessitating ammonia imports from inland renewables-rich provinces. If TTs are used, ammonia fuel demand in the southern coastal regions would primarily be met from nearby areas with CO_2 sequestration sites. In contrast, under the PL scenario, all ammonia fuel demand could be supplied through long-distance pipeline transportation from Inner Mongolia and Qinghai. The difference in transportation radius between the two modes is the main reason for the distinct layouts and trade patterns in the scenarios. Given the spatial mismatch between ammonia fuel supply and demand, the choice of transportation mode is crucial for designing the supply chain in interregional trade.

Pipelines cut costs and emissions across scenarios

Pipeline transportation can lower the accumulated CO_2 emissions and average supply cost of ammonia compared with TTs, as illustrated in Figures 4A and 4B. Implementing pipelines promotes electrolytic hydrogen-based production by enabling economical green ammonia trade from northern regions to southern coastal regions. Consequently, cumulative CO_2 emissions in pipeline scenarios are 3%–13% lower than with TTs across technological and policy sensitivities. From 2030 to 2045, most scenarios with a medium carbon tax (111 USD/t CO_2 by 2060) may see an increase in the average supply cost of ammonia up to around 532 USD/t. By 2060, pipelines can reduce costs to 6%–20% above 2021 levels, whereas TTs may maintain higher cost premiums ranging from 29% to 41% (excluding best and worst). This indicates that pipelines can improve ammonia affordability in the long term.

Pipelines enhance the accessibility of ammonia from low-cost northern production areas to coastal provinces by facilitating long-distance fuel transport, effectively reducing supply costs in coastal regions (Figures 4C and 4D). Although supply costs might slightly increase in some inland provinces under the PL scenario, southeastern coastal provinces could witness a decrease in ammonia supply costs ranging from 12 USD/t to 269 USD/t. The overall cost savings in coastal provinces significantly outweigh the cost increases in inland regions. Comparing PL and no ammonia shipping demand scenario

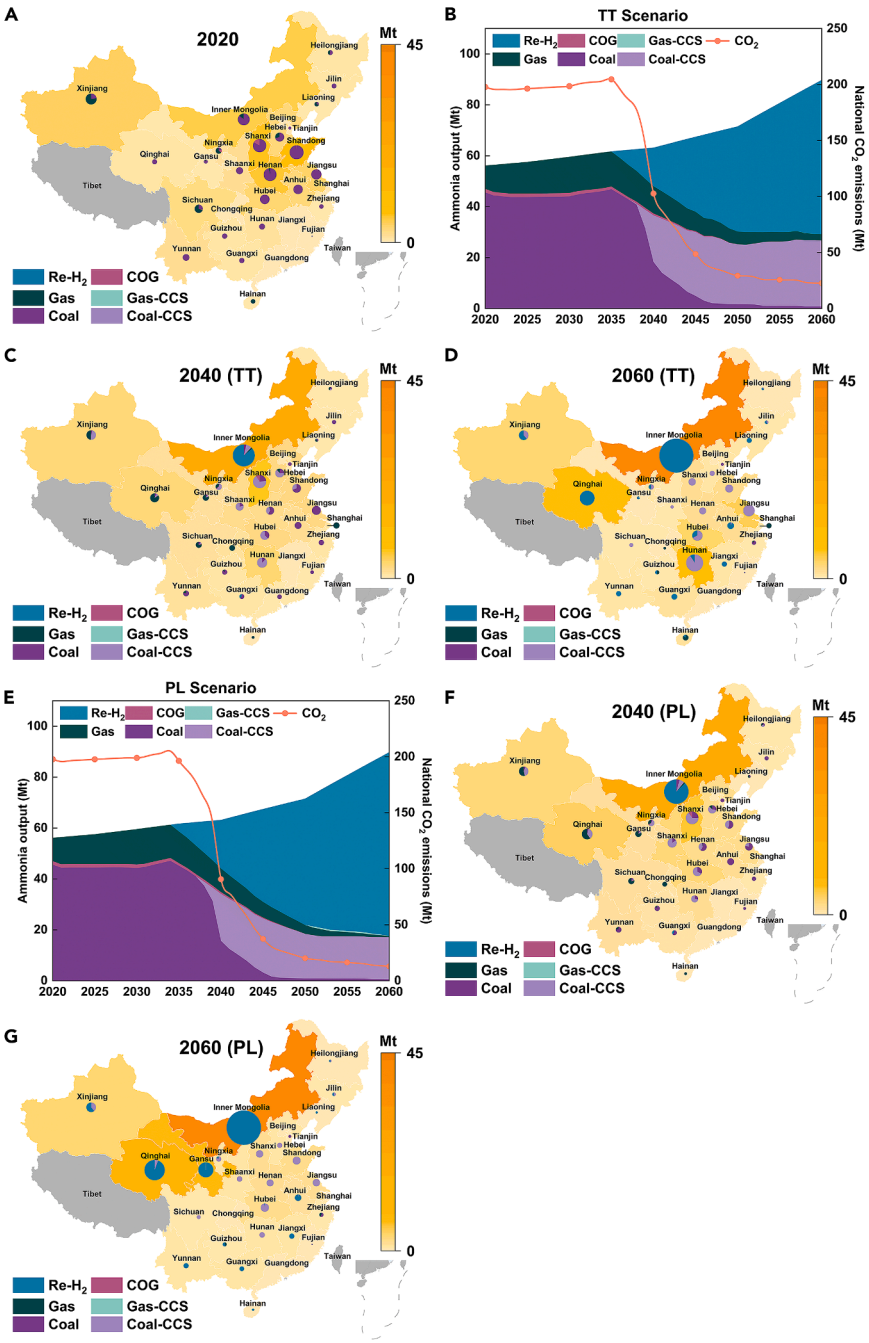


Figure 2. China's ammonia production technology mix evolution and spatial distribution

(A) Provincial ammonia production and technology mix in 2020.

(B and E) National ammonia technology mix pathways and CO_2 emissions under tanker truck (TT) and pipeline (PL) scenarios.

(C and F) Provincial ammonia production and technology mix in 2040 under TT and PL scenarios.

(D and G) Provincial ammonia production and technology mix in 2060 under TT and PL scenarios.

Ammonia production refers to equivalent amounts of ammonia for all commodities produced in a specific region. The base color of the map and the size of the pie charts for each province represent the equivalent ammonia production for that year. The pie charts illustrate the technology mix of each province's production that year. The technology table consists of conventional coal-based (coal), coke oven gas-based (COG), and gas-based (gas) routes; coal-based and gas-based technologies equipped with CCS (coal-CCS and gas-CCS); and electrolytic hydrogen-based ammonia powered by renewable solar and wind (Re-H_2). The hydrogen-based ammonia is exclusively produced from renewable hydrogen, which is generated from water electrolysis using solar/wind power.

(NoFD) shows that emerging fuel demand has a minimal impact on costs and emissions with pipelines, while TT and NoFD reveal considerably higher costs and emissions with trucks.

Advancements and applicability of nascent technologies are vital to controlling cost premiums and reducing cumulative emissions, especially for hydrogen-based technology. Restricted renewable availability could result in a cost premium increase of 14%–34% by 2060, whereas high hydrogen costs may lead to an increase of 20%–41%. Furthermore, restricted renewable availability is expected to contribute to an additional 6%–9% in cumulative emissions, while high hydrogen costs could cause a 9%–15% increase in emissions. However, the most optimistic scenario (best) for future renewable availability and nascent technologies costs could even result in a 10% cost reduction by 2060 compared with 2021 and a 9% reduction in cumulative emissions compared with PL.

Applying a low carbon tax (55 USD/t CO_2 by 2060) could keep ammonia costs moderate under the PL and low carbon price (PL_CP−) scenario, with a maximum increase of 16% by 2045. Although a lower carbon tax might result in a similar technology mix by 2060, varying emission pathways could create significant discrepancies in cumulative environmental impact. The cumulative emissions would be 51% higher than the PL scenario (111 USD/t CO_2 by 2060), which implies a trade-off between decarbonization speed and economic affordability. We also find that under the scenario with a carbon price curve that grows slowly in the early years but accelerates later (PL_CP+), the cumulative carbon emissions are even 6% higher than in the PL scenario. This indicates that increasing the carbon tax in the short term can help the ammonia sector achieve greater cumulative emission reductions, whereas even a high carbon tax in the long term may struggle to further eliminate the sector's residual emissions.

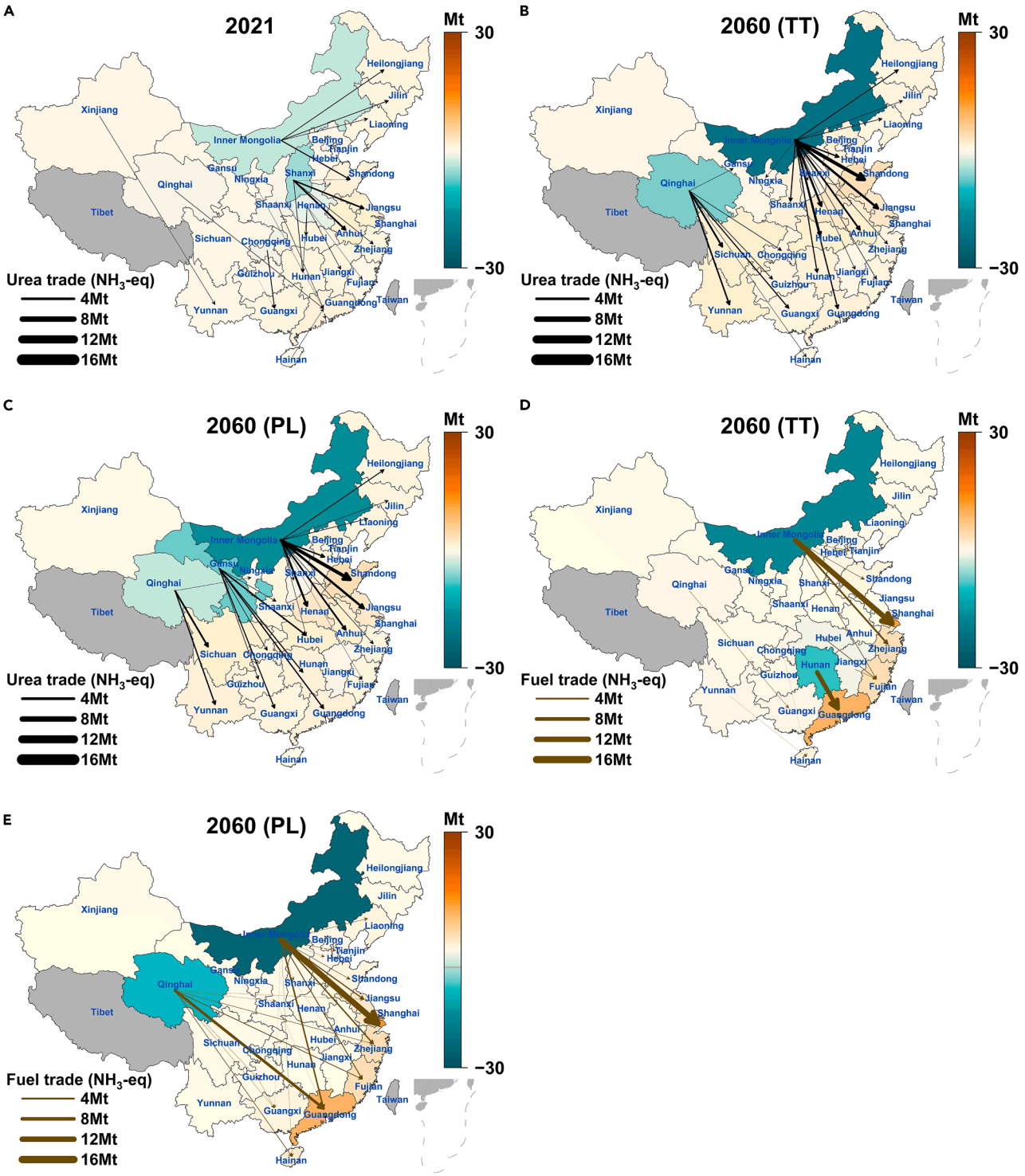


Figure 3. China's ammonia equivalence interprovincial transportation

(A) Interprovincial urea (ammonia equivalence) transportation in 2021.

(B and D) Interprovincial urea and ammonia fuel (ammonia equivalence) transportation in 2060 under the tanker truck (TT) scenario.

(C and E) Interprovincial urea and ammonia fuel (ammonia equivalence) transportation in 2060 under the pipeline (PL) scenario.

Ammonia equivalence interprovincial transportation refers to equivalent amounts of ammonia for all commodities traded from one specific region to another. The base color of the map for each province represents the net import (orange) or export (green) ammonia equivalence for that year. Arrows indicate the direction of trade, and the thickness of the edges represents the trade volume of ammonia equivalence. The results for 2040 are shown in Figure S10.

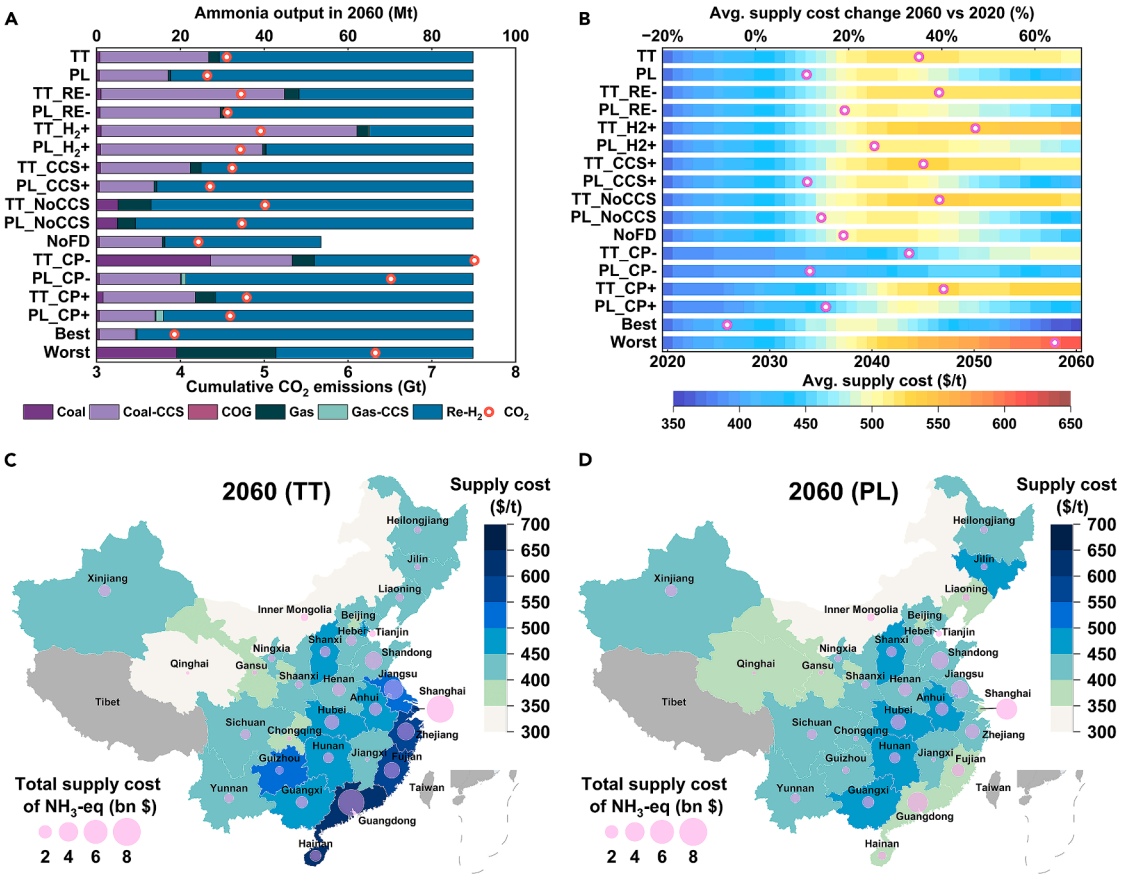


Figure 4. Emission trajectories and cost trends and spatial distribution in different scenarios

(A) National cumulative CO_2 emissions from ammonia production and technology mix in 2060, where Re-H_2 indicates electrolytic hydrogen-based ammonia powered by renewable solar and wind.

(B) National average supply cost trends of ammonia.

(C and D) Provincial unit supply cost and total supply cost of ammonia equivalence in 2060 under the tanker truck (TT) and pipeline (PL) scenario.

The scenarios are grouped by transportation mode, TT and PL, and include the baseline scenarios (TT, PL), limited renewable resource availability (TT_RE−, PL_RE−), high green hydrogen costs (TT_H2+, PL_H2+), high CCS costs (TT_CCS+, PL_CCS+), no CCS availability (TT_NoCCS, PL_NoCCS), no ammonia shipping demand (TT_NoFD, PL_NoFD), low carbon price (TT_CP−, PL_CP−), and exponential carbon price growth (TT_CP+, PL_CP+). In addition, we construct a "best" case with high renewable availability, low green hydrogen and CCS costs, and pipeline transport, and a "worst" case with limited renewables, high green hydrogen and CCS costs, and truck transport. The base color of the map for each province represents the supply cost of ammonia equivalence for that year. The size of each province's bubble represents its total supply cost of ammonia. Supply cost essentially consists of local production cost plus interregional trade cost, with specific calculation methods available in the methods section.

Ammonia fuel needs pipelines and sizable carbon prices

Pipeline transportation can significantly reduce the spatial disparities of ammonia fuel costs (Figure 5A). Under the TT scenario, areas with higher shipping fuel demand tend to have higher marginal costs, with the top four ammonia fuel consumption regions facing marginal costs of 633–699 USD/t (versus an average cost of 373 USD/t in 2020). However, pipeline usage can reduce these marginal costs to 385–398 USD/t by 2060 across provinces.

Reactive nitrogen emissions from ammonia fuel might be sizable in absolute terms yet generally modest relative to national totals (Figure 5B). Under the medium emissions level, ammonia fuel could yield hundreds of kilotons of NH_3, tens of kilotons of N_2O, and over 600 kt of NO_x. In relative terms, except for NO_x, even the high emission level corresponds to <5% of current national totals. Meanwhile, emissions vary by orders of magnitude across emission levels, especially for NO_x. This underscores the need for more accurate, stage-resolved measurements and explicit emission standards to bound uncertainty and secure net air-quality benefits.

Pipeline transportation is essential for making ammonia economically viable as a shipping fuel under a medium carbon tax (Figure 5C). We introduce an "equivalent fuel oil price" for ammonia by converting on the basis of lower heat value and engine efficiency, which enables a direct cost comparison with fuel oil for the same shipping service (see Note S5 for detailed calculations). A marginal cost of 400 USD/t of ammonia corresponds to an equivalent fuel oil price of 1,000 USD/t, which exceeds historical marine fuel oil prices in most periods (400–1,200 USD/t between 2021 and

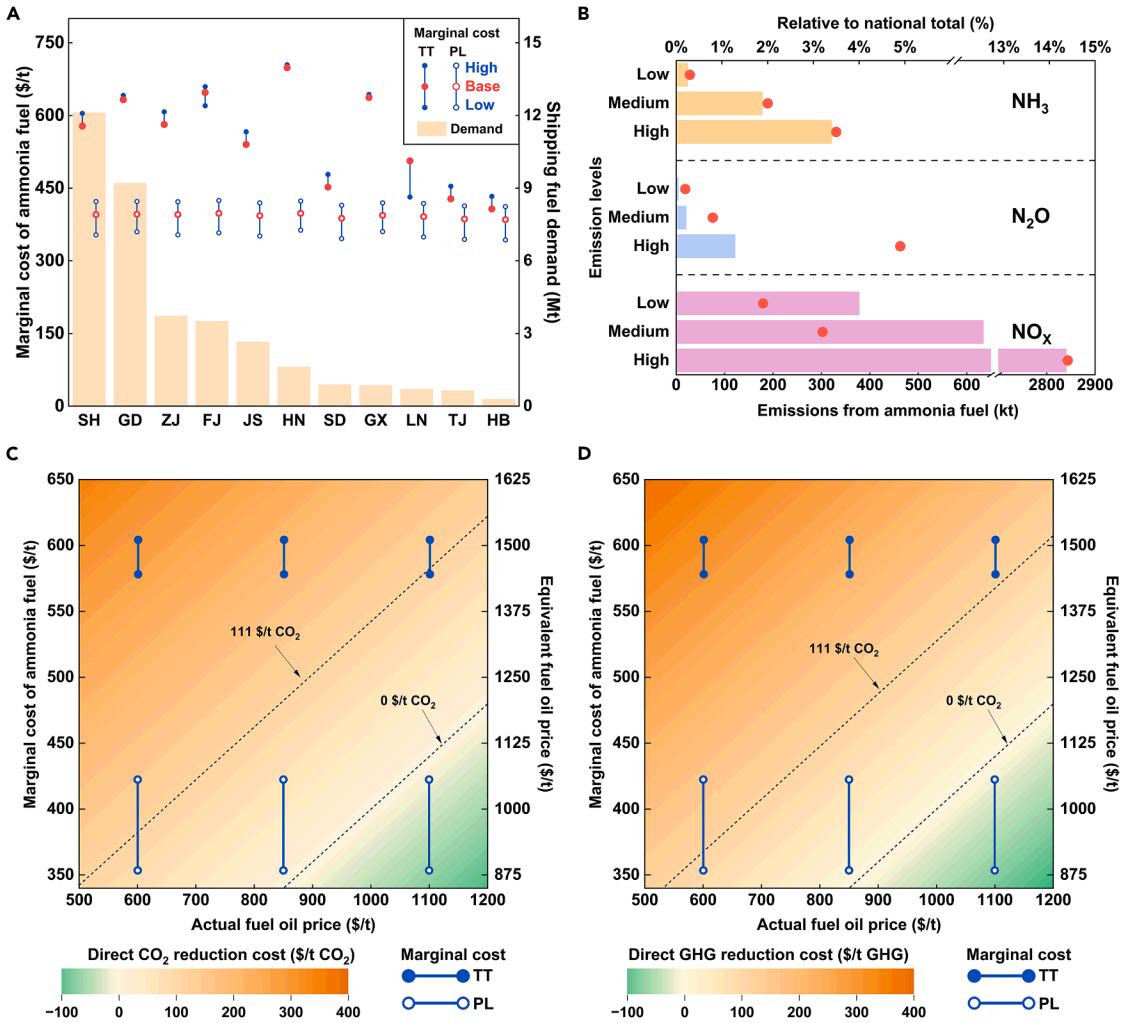


Figure 5. Cost-effectiveness and environmental impact assessments of ammonia fuel

(A) Provincial shipping fuel demand (baseline) and ammonia marginal cost ranges across low and high fuel demand scenarios.

(B) Air pollutants estimation from ammonia fuel. We reference three emission-level scenarios (low/medium/high) for each species (NH_3, N_2O, and NO_x)[36] and report the absolute emissions from ammonia fuel use (bottom *x* axis) and the relative contribution to the national total (top *x* axis). Each colored bar shows emission values, and red dots indicate the ratio relative to the current national total.

(C and D) (C) Direct CO_2 emission and (D) direct GHG emission reduction cost in the shipping sector as a function of the marginal cost of ammonia and price of marine fuel oil in Shanghai.

For scenario name, TT refers to tanker truck and PL means pipeline. Equivalent fuel oil price is calculated by comparing the cost of ammonia to the cost of fuel oil required to provide the same shipping service, considering both heat value and engine efficiency. The price of ammonia fuel is estimated to be equal to the marginal cost of ammonia in 2060. From January 2021 to May 2024, the price of marine fuel oil (0.5% low sulfur) at China's major ports fluctuated between 400 and 1200 USD/t obtained from wind. Excluding the price spike caused by the Russia-Ukraine conflict, the price of oil has generally been around 600 USD/t most of the time. Within this range, three oil price scenarios are selected, corresponding to 600, 850, and 1,100 USD/t, respectively.

2024; most of the time it is around 600 USD/t). The life-cycle emissions reduction cost in shipping varies with fuel oil price uncertainty. Under the PL scenario, ammonia fuel becomes cost-effective if fuel oil prices exceed approximately 600 USD/t, as its emissions reduction cost falls below the 2060 carbon tax of 111 USD/t CO_2. Conversely, using trucks could result in an equivalent fuel oil price of around 1,460 USD/t in some southeastern coastal provinces. For instance, in Shanghai, the break-even CO_2 price for making ammonia competitive with fuel oil would exceed 250 USD/t if fuel oil prices are around 600 USD/t.

The climate impact of replacing fuel oil with ammonia depends jointly on the emissions intensity of the ammonia production route and the magnitude of additional N_2O. As shown in Figures 5C, 5D, and S9, wider color bands indicate greater relative emissions reductions between ammonia fuel and fuel oil, comparing direct CO_2, direct GHG, and life-cycle GHG emissions. Under the medium N_2O emission factor, the direct N_2O emissions from ammonia combustion are 7.8 times higher than those from fuel oil, yet the corresponding reduction in direct GHG emissions relative to fuel oil is only about 10% smaller than the reduction in direct CO_2 emissions. Under the high

N_2O emissions factor, however, the reduction in direct GHG emissions falls to only about 30% of the reduction in direct CO_2 emissions, highlighting the importance of controlling N_2O emissions at the engine level. Life-cycle GHG emissions reductions further depend on the upstream production route. For green-hydrogen-based ammonia, life-cycle GHG reductions are approximately 20% higher than the direct GHG reductions, whereas for coal with CCS routes, life-cycle GHG reductions are only about two-thirds of the corresponding direct GHG reductions. This underscores the critical role of hydrogen-based technologies in lowering overall environmental impacts.

DISCUSSION

This study provides an integrated, supply-chain perspective on how concurrent transformations on the supply and demand sides reshape the technological mix and spatiotemporal deployment of ammonia production in China. We find that green-hydrogen-based technologies begin to deploy at scale around 2030 and become the dominant production route by 2060, supporting more than 90% of emissions reductions across most scenarios. Meanwhile, low-carbon capacity is increasingly concentrated in inland provinces with abundant renewable resources, while emerging maritime fuel demand is clustered in eastern and southeastern coastal regions. The pronounced and persistent spatial mismatch between supply and demand should be bridged by large interregional ammonia flows. Therefore, policymakers need to evaluate and redesign the ammonia supply chain with coordinated planning across production technologies, plant siting, and transport infrastructure to align low-carbon resource endowments with evolving demand centers.

China's ammonia industry needs to scale up nascent low-carbon technologies, especially green-hydrogen pathways, to achieve deep emissions reductions. Accelerating technological advancements in solar panels, wind turbines, battery systems, and electrolyzers could reduce costs and enhance the feasibility of green ammonia production. Several previous analyses have expressed skepticism about the near-term cost-competitiveness of green ammonia.[37] In contrast, our results indicate that green-hydrogen-based routes can become cost-competitive within roughly a decade at a carbon tax of around 30 USD/t CO_2 when hydrogen costs are endogenously optimized against China's projected renewable equipment costs (see Note S6 for detailed assumptions). However, the rapid expansion of electrolytic routes comes with a sizable supply-cost premium: ammonia supply costs rise by about 16%–37% above 2020 levels by 2045 under moderate carbon tax trajectories. Although continued technology progress lowers production costs of hydrogen-based routes, infrastructure choices strongly mediate how supply costs evolve after 2045, since they determine the cost-effectiveness of long-distance green ammonia transport. Pipelines can increase the share of green-hydrogen-based production and allow delivered ammonia costs in 2060 to fall to about 90%–129% of their 2020 levels. By contrast, relying on truck transportation shows little reduction in delivered costs after 2045 despite continued declines in renewable equipment costs. Under a 20% increase in hydrogen costs by 2060, pipelines could sustain an over 50% share of green-hydrogen-based production, whereas truck-based configurations become dominated by coal-CCS (see Note S7 for the sensitivity analysis). Effective carbon-pricing instruments, sustained reductions in hydrogen technology costs, and timely deployment of ammonia pipelines are all critical to unlock large-scale adoption of hydrogen-based ammonia at manageable system costs.

The economics of deploying ammonia as a marine fuel are particularly challenging. Even if green hydrogen were to reach cost parity with gray hydrogen at the point of production, our results indicate that replacing fuel oil with green ammonia would still entail abatement costs of roughly 110 USD/t CO_2. In our scenarios, green ammonia begins to enter the supply mix around 2030, but ammonia fuel becomes competitive with marine fuel oil only by 2060 and only under moderate carbon prices when supported by pipeline transport that lowers delivered costs. The large-scale deployment of ammonia fuel will therefore require substantial additional policy support, such as shipping-sector pricing mechanisms or long-term procurement schemes. The IMO is negotiating a sector-wide GHG pricing mechanism for international shipping, under which a global levy on ship emissions is expected to take effect from 2028 and channel part of the revenues into low-carbon fuels to support the long-term net-zero target in global shipping.[38] However, the recent adjournment of the IMO proposal, reflecting divergent positions among major economies,[39,40] underscores the need for sustained global coordination or, where multi-lateral agreement proves elusive, region-specific measures.

The environmental impacts of ammonia as a marine fuel appear manageable if robust regulation is in place. Compared with fuel oil, using ammonia inevitably increases both direct and indirect N_2O emissions along the nitrogen cycle per unit of shipping service. Nevertheless, in hydrogen-based ammonia pathways, life-cycle GHG emissions per unit of shipping service are only about 5% of those from conventional fuel oil. When scaled to the national level, reactive nitrogen emissions from large-scale ammonia use in shipping remain below roughly 5% of current national totals in most scenarios. Even so, given that the planetary boundary of reactive nitrogen has already been exceeded,[14] we recommend performance-based standards that jointly constrain emissions and leakage along the supply chain, including limits and monitoring for reactive nitrogen emissions and stringent leakage control with regular leak detection and repair.

The China case offers insights that are only partly transferable and should be applied to other regions with caution. Similar spatial mismatches between low-carbon resource hubs and ammonia demand centers are observed in other regions. For example, prospective exporting regions such as Australia and the Middle East might supply emerging import markets in Europe and northeast Asia.[41,42] The integrated, supply-chain-based framework developed here may be useful for other net-zero-seeking economies planning large-scale deployment of low-carbon ammonia. Furthermore, differences in renewable equipment costs and natural gas prices mean that the production and end-use competitiveness of green ammonia will remain highly region specific.[43,44] In China, relatively low renewable costs and a structural dependence on imported oil strengthen the motivation to replace marine fuel oil with domestically produced low-carbon fuels,[45] whereas in gas-rich regions, gray or blue ammonia may remain more cost-competitive for longer unless reinforced by

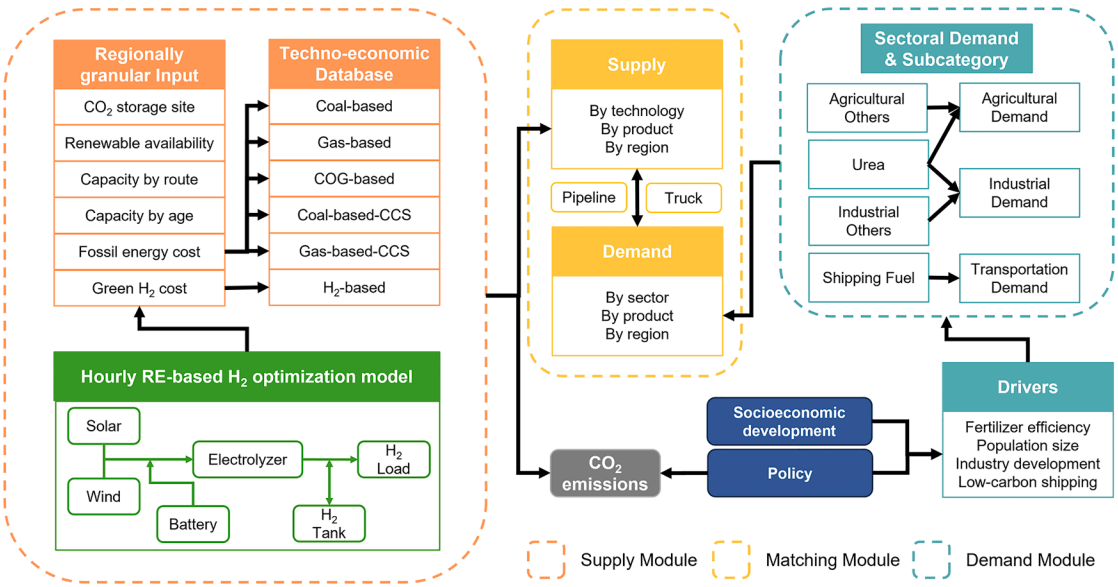


Figure 6. Overview of the integrated modeling framework

The framework combines a regionally granular technoeconomic database, sectoral demand forecast, and different transport modes.

stringent climate policies or dedicated support instruments (see Note S6). Even so, several conclusions from the China case appear broadly applicable to ammonia and shipping decarbonization, including the importance of coherent decarbonization policy support, forward-looking planning of pipeline infrastructure, and sustained support for hydrogen-based technology innovation.

Several limitations point to directions for future work. First, our analysis focuses on domestic supply chains and shipping fuel demand without explicitly considering the IMO's net-zero target. Coupling China's ammonia sector to global shipping markets would provide a more complete picture of trade and competitiveness impacts. We also do not explicitly consider other emerging end uses of ammonia (e.g., ammonia-coal cofiring and long-duration energy storage), and we simplify downstream product representation by treating urea as the main interregional ammonia-derived product. Second, technology choices for marine fuels remain highly uncertain. Ammonia as a shipping fuel faces several challenges, including high toxicity, combustion stability issues, and the potential for nitrogen oxide emissions.[46,47] Meanwhile, ammonia-fueled engines and fuel-supply systems are still in the research and development stage.[48] Ammonia also competes with alternative fuels such as methanol, hydrogen, and biodiesel in the shipping sector.[49] Future work should therefore undertake comparative assessments across multiple fuel candidates, examining technological feasibility, economic performance, environmental impacts, and appropriate application niches. Third, while we explore ranges of emission factors based on engineering estimates for ammonia fuel,[50] more experimental measurements across production, transport, and combustion stages are needed to refine the accounting of reactive nitrogen emissions and to evaluate localized air quality impacts. Finally, we did not assess ecosystem impacts: production hubs may face water and land-use constraints, and poorly planned renewable build-outs and associated corridors can fragment habitats. Future work should adopt a screen and then optimize capacity siting strategy that excludes protected and key biodiversity areas and manages water use to respect ecological-flow requirements.

METHODS

This study develops a planning model that integrates a bottom-up demand forecast and an optimization-based supply module, as shown in Figure 6. It covers the entire ammonia industry chain, from raw material production and critical downstream products to interregional transportation. Sector-specific ammonia demands can be predicted based on various drivers. The entire supply chain, including technology selection, capacity allocation, and interregional trade, is optimized with minimum cost. The exogenous carbon tax is the driving factor that simulates long-term decarbonization pathways. With provincial-level spatial resolution, this model accounts for resource endowment differences across regions. For example, this study develops a technology-rich technoeconomic analysis database for ammonia production technologies with a provincial resolution. Based on this database, a multi-regional planning model is developed with an annual temporal resolution from 2021 to 2060, taking 2020 as the base year. Due to data availability, 30 provinces within mainland China are included, excluding Hong Kong, Macau, Taiwan, and Tibet. The detailed methods, key input data, and emissions accounting are listed in Note S8.

Technoeconomic database

The technoeconomic database covers the energy consumption, investment costs, operation and maintenance (O&M) costs, carbon emissions, and regional-specific energy costs for each technology. The costs of provincial fossil fuels are sourced from the Wind and CEIC Data Global Database. The majority of ammonia

synthesis plants based on fossil fuels operate almost continuously, with a high capacity factor, benefiting from the mature and stable supply system of fossil energy. However, the hydrogen-based technologies considered in this study are derived from wind and solar generation systems. The variability in wind and solar output may lead to instability in hydrogen production, posing technical challenges for ammonia synthesis. This study assumes that the Haber-Bosch process in the ammonia synthesis system requires continuous operation and a stable hydrogen supply. Therefore, we have developed an hourly model to explore the optimal configuration of the hydrogen system with fixed hourly hydrogen demand. Hydrogen is produced by water electrolysis in alkaline electrolyzers powered by wind and solar generation. The lithium-ion batteries and hydrogen storage tanks are equipped to ensure a stable hydrogen supply for the ammonia plant. As the wind and solar power output curves differ by region, the optimal configuration of power capacity, electrolyzer capacity, lithium-ion battery capacity, and hydrogen storage tank capacity also varies by region. Additionally, the future cost reduction trends of wind and solar power, electrolyzers, and storage systems are considered due to technological advancements. The optimal configuration in the same region can be different across different years. The wind and solar profiles were obtained from http://renewable.ninja/ using the 2016 weather year.

The levelized cost of hydrogen (LCOH) is defined as the annualized cost of the system divided by the annual hydrogen production. The objective function is to minimize the annualized cost of the system, where lithium-ion batteries are categorized into power capacity and energy capacity costs (Equation 1). The wind and solar output are less than the installed capacity multiplied by the maximum output curve of the unit installation at a certain time slice (Equations 2 and 3). The generation from wind and solar power, plus the net discharge from batteries, is used to power the electrolyzer, with the power being less than the electrolyzer capacity (Equations 4 and 5). $lcoh_{r,t}$ is the LCOH in province r and year t, and $y_{r,t,solar}$ and $y_{r,t,wind}$ refer to the installed capacities of solar and wind power, respectively. $z_{r,t}$ is the electrolyzer capacity, and $u_{r,t}$ and $v_{r,t}$ represent the battery power and energy capacities. $w_{r,t}$ denotes the hydrogen tank capacity. $AC_{t,solar}$, $AC_{t,wind}$, $AC_{t,elec}$, $AC_{t,batpower}$, $AC_{t,batenergy}$, and $AC_{t,tank}$ refer to the annualized costs per unit capacity of solar, wind, electrolyzer, battery power, battery energy, and hydrogen tank in year t, respectively. $HD_{h,r,t}$ is the fixed hydrogen hourly demand. $x_{h,r,t,solar}$ and $x_{h,r,t,wind}$ denote the solar and wind generation in hour h. $MAXGEN_{h,r,t,solar}$ and $MAXGEN_{h,r,t,wind}$ are the maximum hourly generation factors. $x^{discharge}_{h,r,t,bat}$ and $x^{charge}_{h,r,t,bat}$ are the battery discharging and charging amounts, respectively. $x_{h,r,t,hydrogen}$ is the hydrogen output in hour h. η_{p2h} represents the electricity-to-hydrogen conversion efficiency:

$$lcoh_{r,t} = \left(y_{r,t,solar}\cdot AC_{t,solar} + y_{r,t,wind}\cdot AC_{t,wind} + z_{r,t}\cdot AC_{t,elec} + u_{r,t}\cdot AC^{power}_{t,bat} + v_{r,t}\cdot AC^{energy}_{t,bat} + w_{r,t}\cdot AC_{t,tank}\right) \Big/ \sum_{h} HD_{h,r,t}, \quad \text{(Equation 1)}$$

$$x_{h,r,t,solar} \le y_{r,t,solar}\cdot MAXGEN_{h,r,t,solar}, \quad \text{(Equation 2)}$$

$$x_{h,r,t,wind} \le y_{r,t,wind}\cdot MAXGEN_{h,r,t,wind}, \quad \text{(Equation 3)}$$

$$x_{h,r,t,wind} + x_{h,r,t,solar} + x^{discharge}_{h,r,t,bat} - x^{charge}_{h,r,t,bat} \le z_{r,t}, \quad \text{(Equation 4)}$$

$$\left(x_{h,r,t,wind} + x_{h,r,t,solar} + x^{discharge}_{h,r,t,bat} - x^{charge}_{h,r,t,bat}\right)\cdot \eta_{p2h} = x_{h,r,t,hydrogen}. \quad \text{(Equation 5)}$$

Lithium-ion batteries and hydrogen storage tanks need to meet a series of physical constraints. The state of charge for each time slice equals the state of charge from the previous time slice plus energy input minus energy output (Equations 6 and 9). The output of hydrogen used for ammonia production is fixed. The state of charge of the storage must be less than the energy capacity of the storage device (Equations 7 and 10). For batteries, the charging and discharging power must be less than the battery's power capacity (Equation 8). $l_{h,r,t,bat}$ is the state of charge of battery in hour h, province r, and year t. $x^{discharge}_{h,r,t,bat}$ and $x^{charge}_{h,r,t,bat}$ denote battery discharge and charge amounts, respectively. $eff_{discharge}$ and eff_{charge} are the efficiencies of battery discharge and charge processes. $v_{r,t}$ represents the battery energy capacity. $u_{r,t}$ is the battery power capacity. $l_{h,r,t,tank}$ is the state of charge of the hydrogen tank in hour h. $x_{h,r,t,hydrogen}$ denotes hydrogen production in hour h. $HD_{h,r,t}$ is the fixed hydrogen hourly demand. $w_{r,t}$ denotes the hydrogen tank capacity:

$$l_{h,r,t,bat} = l_{h,r,t-1,bat} - x^{discharge}_{h,r,t,bat} \Big/ eff_{discharge} + x^{charge}_{h,r,t,bat} \times eff_{charge}, \quad \text{(Equation 6)}$$

$$l_{h,r,t,bat} \le v_{r,t}, \quad \text{(Equation 7)}$$

$$x^{discharge}_{h,r,t,bat} + x^{charge}_{h,r,t,bat} \le u_{r,t}, \quad \text{(Equation 8)}$$

$$l_{h,r,t,tank} = l_{h,r,t-1,tank} + x_{h,r,t,hydrogen} - HD_{h,r,t}, \quad \text{(Equation 9)}$$

$$l_{h,r,t,tank} \le w_{r,t}. \quad \text{(Equation 10)}$$

For CCS technology, the entire process of CO_2 capture, transport, and storage is assessed for cost calculation. The foresight into the full-cycle cost of CCS indicates a downward trend in the future attributed to technological progress, which is obtained from the literature.[51] For electricity, we consider the heterogeneity of regional grid emission factors and project the annual emission factors for each province by 2060. The provincial grid power emission factors are obtained from Cai et al.[52] The emission factor projections are based on the CO_2 trajectory of China's power sector under the carbon neutrality scenario[53] and assume the same pace across provinces.

Demand module

Ammonia demand is categorized into three sectors: agriculture, industry, and transportation. An estimation method is developed for base-year data to depict the heterogeneity of ammonia demand structure for each province. As a key downstream material of ammonia, urea is utilized in both industrial and agricultural

sectors and is widely traded across regions. Therefore, agricultural and industrial ammonia consumption are further disaggregated into urea consumption and non-urea consumption. Industrial ammonia use, agricultural ammonia use, industrial urea consumption, and agricultural urea consumption at the national level can be obtained from the literature.[54,55] Provincial ammonia production, urea production, fertilizer consumption, and urea consumption can also be sourced from the literature and yearbooks.[55,56] This study estimates provincial agricultural ammonia consumption via nitrogen and compound fertilizer consumption. A uniform fertilizer product structure is assumed for each province. The formulas for calculating agricultural urea and other agricultural ammonia demand are provided (Equations 11 and 12). For example, the agricultural urea demand is calculated as the sum of nitrogen fertilizer and compound fertilizer quantities, each multiplied by the respective proportion of urea they contain. The parameters can be obtained from the national consumption structure. Next, we calculate each province's industrial urea demand and other industrial ammonia demands. Industrial urea consumption is derived by subtracting agricultural usage from total provincial data (Equation 13). Other industrial demands are calculated by deducting urea production and agricultural needs from total ammonia production (Equation 14). Results show the proportions of industrial and agricultural ammonia consumption across different provinces, as well as the provincial urea demand. Finally, the data provided in Figure 1 indicate the ammonia equivalence consumption for all commodities. For example, the ammonia consumption from agricultural urea is equal to agricultural urea demand multiplied by the stoichiometry between ammonia and urea. $AUD_{r,2020}$ represents agricultural urea demand in province r in 2020 (equivalent ammonia), a and b are parameters representing nitrogen and compound fertilizer structure, and $NFD_{r,2020}$ and $CFD_{r,2020}$ denote nitrogen fertilizer and compound fertilizer consumption in 2020. $AAOD_{r,2020}$ is the agricultural ammonia non-urea demand. c and d are parameters for the non-urea fertilizer structure. $IUD_{r,2020}$ denotes industrial urea demand, $UD_{r,2020}$ is total urea demand, $IAOD_{r,2020}$ is industrial ammonia non-urea demand, and $AP_{r,2020}$ and $UP_{r,2020}$ are total ammonia and urea production in 2020, respectively:

$$AUD_{r,2020} = a \cdot NFD_{r,2020} + b \cdot CFD_{r,2020}, \quad \text{(Equation 11)}$$

$$AAOD_{r,2020} = c \cdot NFD_{r,2020} + d \cdot CFD_{r,2020}, \quad \text{(Equation 12)}$$

$$IUD_{r,2020} = UD_{r,2020} - AUD_{r,2020}, \quad \text{(Equation 13)}$$

$$IAOD_{r,2020} = AP_{r,2020} - UP_{r,2020} - AAOD_{r,2020}. \quad \text{(Equation 14)}$$

Future projections for agriculture and industry follow consistent provincial change rates, which are obtained from the literature.[57] Agricultural demand is estimated to decline with the improvement of fertilizer efficiency and shrinkage of the population.[58] Industrial demand is expected to grow moderately in line with the development of related industries.[59] Given the different ammonia consumption structures across provinces, the future trend of total traditional ammonia consumption in each province will not be uniform. Transportation ammonia demand is closely linked to shipping energy demand. The provincial ammonia fuel demand is estimated by multiplying activity level, ammonia fuel contribution ratio, and energy efficiency. Provincial maritime freight turnover in the base year, forecasts of maritime freight turnover, and the annual contribution ratio of ammonia fuel are all obtained from the literature or statistical yearbooks.[56,60] We assume that the maritime freight turnover in each province will increase at the same rate as the national average.

Supply module

The supply module evaluates the technological revolution of ammonia production processes and interregional trade patterns based on an optimization approach. This study considers five ammonia production technologies: coal-based, gas-based, coal with CCS, gas with CCS, and green-hydrogen-based routes. The model incorporates the technoeconomic analysis database and provincial resource and industry status. Interregional trade is delineated to balance the supply and demand of each province. This study assumes urea is the primary transported product of conventional demand, neglecting the interregional transport of other products. Future direct interregional transport of ammonia fuel may occur, given its use as a final product in the shipping sector. The reasons are as follows: (1) urea is the primary downstream product of ammonia, accounting for approximately 70% of conventional ammonia demand. Additionally, urea is extensively traded across provinces in the commodity market.[61] (2) Some downstream products, like ammonium nitrate, are prone to explosion and decomposition, making them unsuitable for long-distance transport.[62] (3) Chemical products such as melamine, cyanuric acid, and urea-formaldehyde resins are intermediates requiring further downstream processing to yield final products.[63] The module can output the annual provincial technology mix of ammonia production during the planning period and interprovincial transport of ammonia and urea. The capacity, energy consumption, and carbon emissions can also be accounted for based on the results.

The objective function is to minimize the total discounted cost of the supply costs of ammonia and key downstream products. It encompasses various components such as investment, retrofitting, operation and maintenance, energy consumption, CO_2 transport and storage, product transportation, and emissions (Equations 15, 16, 17, 18, 19, 20, 21, and 22). Investment costs represent the capital expenditure related to plants. Retrofitting costs refer to installing carbon capture facilities in pre-existing fossil-fuel-based plants. Energy expenditure consists of fossil fuel costs, hydrogen costs, and electricity costs incurred during the ammonia production process. Hydrogen costs are calculated by summing the products of each project's LCOH and its hydrogen production for the accounting year. CO_2 transport and storage costs denote further processing expenses after capturing CO_2. Transportation cost refers to the interprovincial transportation costs of urea and ammonia fuel. The transportation costs of urea and ammonia tankers are proportional to the distance and volume. The construction cost of pipelines is detailed in the pipeline constraints section. The emissions cost is determined by multiplying the net CO_2 emissions by the exogenous carbon tax. The emissions cost is the driving force for adopting nascent technologies. Although the synthesis of other products downstream of ammonia may also involve other costs,

it is not included in the optimization objective. This is because, given a specific demand structure, the energy consumption and emissions associated with synthesizing downstream products from ammonia are fixed. *atc* is the total discounted system cost from 2021 to 2060. $tinv_{r,t}$ represents total investment cost in province *r* in year *t*. $trc_{r,t}$ is the total retrofit cost; $tom_{r,t}$ denotes the total operation and maintenance cost; $tec_{r,t}$ is the total energy cost; $ctsc_{r,t}$ refers to CO_2 transport and storage cost; $ctc_{r,t}$ is the total CO_2 emission tax cost; $ttc_{r,r',t}$ represents the total transport and pipeline cost between provinces *r* and *r*′. *I* is the discount rate; $CAP_{t,p}$ is the unit investment cost of technology *p* in year *t*; $nb_{r,t,p}$ is the newly built capacity; $RC_{t,p}$ is the unit retrofit cost; $rc^{t'}_{r,t,p}$ is the capacity built in *t* and retrofitted in *t*′; $\mu_{t,p}$ is the O&M cost per unit; $ic_{r,t,p}$ is the existing capacity of technology *p*. $FP_{f,r,t}$ is the fuel price; $ao_{r,t,p}$ is the ammonia output; $FU_{t,p,f}$ is the fuel consumption factor; $HP_{r,t}$ is the hydrogen price; $ho_{r,t',t}$ is the hydrogen output from systems built in *t*′ used in year *t*; $PP_{r,t}$ is the electricity price; $PU_{t,p}$ is the electricity consumption; CTS_t is the CO_2 transport and storage unit cost; $cc_{r,t}$ is the captured CO_2; TC_{af} and TC_u are transport cost coefficients for ammonia fuel and urea; $aft_{r,r',t}$ and $ut_{r,r',t}$ are transported quantities; $DIS_{r,r'}$ is the distance between provinces; *ZZ* is the tortuosity factor; $tpc_{r,r',t}$ is the pipeline cost; $em_{r,t}$ is CO_2 emissions; TAX_t is the carbon tax:

$$atc = \sum_{t=2021}^{2060} \sum_{r} \left(tinv_{r,t} + trc_{r,t} + tom_{r,t} + tec_{r,t} + ctsc_{r,t} + ctc_{r,t} + \sum_{r'} ttc_{r,r',t} \right) \Big/ (1+I)^{t-2020}, \quad \text{(Equation 15)}$$

$$tinv_{r,t} = \sum_{p} \sum_{t'=t-TLT_p+1}^{t} \left(CAP_{t',p} \cdot nb_{r,t',p} \cdot I \cdot (1+I)^{-1} / 1 - (1+I)^{-TLT_p} \right), \quad \text{(Equation 16)}$$

$$trc_{r,t} = \sum_{p} \sum_{t'=t-TLT_p+1}^{t-1} \sum_{t''=t'+1}^{t} \left(RC_{t'',p} \cdot rc^{t''}_{r,t',p} \cdot I \cdot (1+I)^{-1} / 1 - (1+I)^{-TLT_p+t''-t'} \right), \quad \text{(Equation 17)}$$

$$tom_{r,t} = \sum_{p} \mu_{t,p} \cdot ic_{r,t,p}, \quad \text{(Equation 18)}$$

$$tec_{r,t} = \sum_{p} \left(\sum_{f \neq hydrogen} FP_{f,r,t} \cdot ao_{r,t,p} \cdot FU_{t,p,f} + \sum_{t'=t-TLT_h+1}^{t} HP_{r,t'} \cdot ho_{r,t',t} + PP_{r,t} \cdot ao_{r,t,p} \cdot PU_{t,p} \right), \quad \text{(Equation 19)}$$

$$ctsc_{r,t} = CTS_t \cdot cc_{r,t}, \quad \text{(Equation 20)}$$

$$ttc_{r,r',t} = \left(TC_{af} \cdot aft_{r,r',t} + TC_u \cdot ut_{r,r',t} \right) \cdot DIS_{r,r'} \cdot ZZ + tpc_{r,r',t}, \quad \text{(Equation 21)}$$

$$ctc_{r,t} = em_{r,t} \cdot TAX_t. \quad \text{(Equation 22)}$$

Hydrogen-based routes exclusively consider renewable energy for green hydrogen, assuming local consumption due to high hydrogen transportation costs. The interprovincial CO_2 transport is ignored due to high transport costs. The source-sink distance within a province is assumed to be 100 km.

Emissions calculation

The emission accounting scope includes direct emissions from the production process and indirect emissions from power generation (Equation 23). The direct carbon emission factor per unit of ammonia production in the first item already accounts for carbon removal:

$$em_{r,t} = \sum_{p} ao_{r,t,p} \cdot \left(EFF_{f,t,p} + PU_{t,p} \cdot EFP_{t,r} \right). \quad \text{(Equation 23)}$$

Supply and demand constraints

Supply and demand constraints state that the annual product demand in each province must be met by production from all technological routes and net inflow (where applicable). Equations 25, 26, and 27 represent the balance for urea, ammonia fuel, and other ammonia demands, respectively. According to our model's framework, we assume that urea and ammonia fuel can be transported across provinces. In contrast, other ammonia demands are subject to integrated production constraints, necessitating self-sufficiency in each province. Equation 28 illustrates that the total ammonia output in each province is allocated to urea, ammonia fuel, and other demands. This signifies that the unit of measure for all demand categories is expressed as equivalent ammonia requirements rather than in physical quantities. $uo_{r,t,p}$ refers to urea output using technology *p* in province *r* in year *t*. $ut_{r,r',t}$ is the urea transported from province *r* to *r*′ in year *t*. $afo_{r,t,p}$ refers to ammonia fuel output. $aft_{r,r',t}$ is ammonia fuel transported via TT from province *r* to *r*′. $afp_{r,r',t}$ is ammonia fuel transported via pipelines from province *r* to *r*′. $aoo_{r,t,p}$ refers to ammonia output for other demands:

$$\sum_{p} uo_{r,t,p} - \sum_{r'} ut_{r,r',t} + \sum_{r'} ut_{r',r,t} = UD_{r,t}, \quad \text{(Equation 25)}$$

$$\sum_{p} afo_{r,t,p} - \sum_{r'} aft_{r,r',t} + \sum_{r'} aft_{r',r,t} - \sum_{r'} afp_{r,r',t} + \sum_{r'} afp_{r',r,t} = AFD_{r,t}, \quad \text{(Equation 26)}$$

$$\sum_{p} aoo_{r,t,p} = AOD_{r,t}, \quad \text{(Equation 27)}$$

$$ao_{r,t,p} = uo_{r,t,p} + afo_{r,t,p} + aoo_{r,t,p}. \quad \text{(Equation 28)}$$

Capacity constraints

For ammonia capacity, it is necessary to comply with constraints such as the annual newly built capacity, decommissioning or early retirement, and maintaining maximum and minimum capacity utilization rates. The capacity calculations for

fossil-based, fossil-based with CCS, and hydrogen-based plants are illustrated in Equations 29, 30, and 31. Considering technological transition and overcapacity in China, some capacities may be decommissioned before the end of their lifespan. The production of each technology cannot exceed capacity and must be above the minimum capacity utilization rate (Equation 32). The annual maximum new capacity constraints are set for a given technological route at both the provincial and the national levels (Equations 33 and 34). *CFL* is the minimum capacity utilization rate. $nbMAX_{r,t,p}$ and $tnbMAX_{t,p}$ denote the maximum provincial and national new capacity additions:

$$ic_{r,t,p} = \sum_{t'=t-TLT_p+1}^{t} nb_{r,t',p} - \sum_{t'=t-TLT_p+1}^{t} \sum_{t''=t'+1}^{t} \left(er_{r,t',p}^{t''} + rc_{r,t',p}^{t''} \right), \quad \text{(Equation 29)}$$

$$ic_{r,t,p} = \sum_{t'=t-TLT_p+1}^{t} nb_{r,t',p} - \sum_{t'=t-TLT_p+1}^{t} \sum_{t''=t'+1}^{t} \left(er_{r,t',p}^{t''} - rc_{r,t',p}^{t''} \right), \quad \text{(Equation 30)}$$

$$ic_{r,t,p} = \sum_{t'=t-TLT_p+1}^{t} nb_{r,t',p} - \sum_{t'=t-TLT_p+1}^{t} \sum_{t''=t'+1}^{t} er_{r,t',p}^{t''}, \quad \text{(Equation 31)}$$

$$ic_{r,t,p} \times CFL \le ao_{r,t,p} \le ic_{r,t,p}, \quad \text{(Equation 32)}$$

$$nb_{r,t,p} \le nbMAX_{r,t,p}, \quad \text{(Equation 33)}$$

$$\sum_{r} nb_{r,t,p} \le tnbMAX_{t,p}. \quad \text{(Equation 34)}$$

Nascent technology constraints

For hydrogen-based technology, hydrogen production should be equal to hydrogen consumption in each zone (Equation 35). The renewable resource availability creates an upper limit for hydrogen production from each province (Equation 36). The estimation of the maximum potential for hydrogen production is related to the wind and solar power potential and projected electricity demand of each province. For CCS, an annual maximum sequestration amount for each region is defined based on the maximum storage potential of each province (Equation 37). *HU* denotes hydrogen consumption per unit ammonia. $MAXH_r$ is the upper limit of green hydrogen availability. $cc_{r,t}$ is CO_2 storage in province *r* in year *t*, and $MAXCCS_{r,t}$ is its upper bound:

$$\sum_{t'=t-TLT_h+1}^{t} ho_{r,t',t} = ao_{r,t,hydrogen} \cdot HU, \quad \text{(Equation 35)}$$

$$ao_{r,t,hydrogen} \cdot HU \le MAX\,H_r, \quad \text{(Equation 36)}$$

$$cc_{r,t} \le MAXCCS_{r,t}. \quad \text{(Equation 37)}$$

Transportation constraints

We assume ammonia is transported in its liquefied form, as ammonia can be easily liquefied under moderate pressure and ambient temperature, which also confers it a cost advantage over hydrogen transport. For truck transport, a simplified integrated cost per ton-kilometer is adopted. For pipelines, we include capital expenditure on infrastructure (unit costs from the literature), fixed O&M, and electricity for pumping. Converting to a per-ton-kilometer basis for ammonia transport, the pipeline cost is roughly an order of magnitude lower than trucking.

The pipeline capacity is calculated as the sum of all newly built pipelines within their lifetimes (Equation 38). The annual ammonia transported through the pipeline must be less than the capacity multiplied by the maximum load factor (Equation 39). Physically, the pipeline from location A to location B is the same as the pipeline from location B to location A, so the cost calculation needs to be divided by 2 to avoid double counting (Equations 40 and 41). $pc_{r,r',t}$ is pipeline capacity. $nbp_{r,r',t}$ denotes newly built pipelines, which are symmetric across *r* and *r*′:

$$\sum_{t'=t-TLT_{pi}+1}^{t} nbp_{r,r',t'} = pc_{r,r',t}, \quad \text{(Equation 38)}$$

$$afp_{r,r',t} \le pc_{r,r',t} \cdot PCF, \quad \text{(Equation 39)}$$

$$nbp_{r,r',t} = nbp_{r',r,t}, \quad \text{(Equation 40)}$$

$$tpc_{r,r',t} = \sum_{t'=t-TLT_{pi}+1}^{t} \left(PCAP_{t'} \cdot nbp_{r,r',t'} \cdot I \cdot (1+I)^{-1} / 1 - (1+I)^{-TLT_{pi}} \right) \Big/ 2 \cdot DIS_{r,r'} \cdot ZZ. \quad \text{(Equation 41)}$$

Costs calculation

The production and supply costs can be calculated based on the results of the planning model. The production cost is defined as the LCOA for producing ammonia within a province each year (Equation 42). Compared with the investment cost in the planning model, production cost calculation considers the annualized investment cost for the capacity of that specific year (Equation 43), which can eliminate the influence of prematurely retired capacities on cost accounting. All other cost components can be directly derived from the results of the planning model. $procost_{r,t}$ is the unit ammonia production cost. $supcost_{r,t}$ is the supply cost, incorporating production cost, import/export adjustments, and transport costs. $imcost_{r,t}$, $excost_{r,t}$, $trcost_{r,t}$, and $pipcost_{r,t}$ denote import, export, truck, and pipeline-related costs, respectively:

$$procost_{r,t} = \left(tinvt_t^n + trc_{r,t} + tom_{r,t} + tec_{r,t} + ctsc_{r,t} + ctc_{r,t} \right) \Big/ \sum_{p} ao_{r,t,p}, \quad \text{(Equation 42)}$$

$$tinvt_i^n = \sum_p \left(CAP_{t',p} \cdot ic_{r,t',p} \cdot I \cdot (1+I)^{-1} / 1 - (1+I)^{-TLT_p} \right).$$

(Equation 43)

The supply cost is defined as the average total cost per ton of ammonia required to meet the ammonia demand of a province each year. The supply cost needs to adjust for the impact of imports and exports on the province's total production cost, add the related transportation costs, and consider the conversion and fugitive losses along the supply chain (Equations 44, 45, 46, 47, and 48). The import cost is equal to the import volume multiplied by the production cost at production place, while the export cost is equal to the export volume multiplied by the local production cost. Transportation costs are accounted for only in the importing province. For example, the annualized cost of the pipeline for that year is borne solely by the importer of the ammonia:

$$supcost_{r,t} = \left(procost_{r,t} \cdot \sum_p ao_{r,t,p} + imcost_{r,t} - excost_{r,t} + trcost_{r,t} + pipcost_{r,t} \right) \Big/ \left(UD_{r,t} + AFD_{r,t} + AOD_{r,t} \right) \cdot (1 - loss),$$

(Equation 44)

$$imcost_{r,t} = \sum_{r'} \left(ut_{r',r,t} + aft_{r',r,t} + afp_{r',r,t} \right) \cdot procost_{r',t},$$

(Equation 45)

$$excost_{r,t} = \sum_{r'} \left(ut_{r,r',t} + aft_{r,r',t} + afp_{r,r',t} \right) \cdot procost_{r,t},$$

(Equation 46)

$$trcost_{r,t} = \sum_{r'} \left(TC_{af} \cdot aft_{r',r,t} + TC_u \cdot ut_{r',r,t} \right) \cdot DIS_{r,r'} \cdot ZZ,$$

(Equation 47)

$$pipcost_{r,t} = \sum_{r'} tpc_{r',r,t} \cdot 2, if \left(afp_{r',r,t} > 0 \right).$$

(Equation 48)

Environmental impact

We quantify the environmental impacts associated with ammonia use as a marine fuel, focusing on emissions of reactive nitrogen species. Specifically, we estimate NO_x, N_2O, and NH_3 leakage under multiple emission-factor scenarios. For comparison, we also evaluate the pollutant profiles of conventional fuel oil, including N_2O, NO_x, and SO_x. In addition to direct emissions, we assess the indirect formation of N_2O arising from unreacted NH_3 and NO_x within the combustion cycle to provide a more complete estimate of the overall climate implications of ammonia-fueled shipping. Detailed assessment of environmental impacts is provided in Note S9.

RESOURCE AVAILABILITY

Lead contact

Please contact the lead contact, Dr. Tianduo Peng (ptd@mail.tsinghua.edu.cn), for information related to the data and model used in this study.

Materials availability

This study did not generate new unique materials.

Data and code availability

The original contributions presented in the study are available at Zenodo: https://doi.org/10.5281/zenodo.16283863. These include the models, input data, and visualization data of this study. Further inquiries may be directed to the lead contact.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.D.J., Z.L., Y.H., and T.P.; methodology, Y.H. and T.P.; software, Y.H.; validation, Y.H. and Z.W.; formal analysis, Y.H. and Z.W.; data curation, Y.H. and Z.W.; visualization, Y.H. and Z.W.; writing – original draft, Y.H. and Z.W.; writing – review & editing, J.D.J., T.P., and Z.L.; funding acquisition, Z.L. and T.P.; supervision, J.D.J., Z.L., and T.P.

DECLARATION OF INTERESTS

J.D.J. is co-founder and chief technology officer of Firma Power LLC, which structures and supplies portfolios of clean electricity to large power consumers. He also serves as an advisor to Eavor Technologies, Rondo Energy, Dig Energy, Karman Industries, Emerald AI, Resilient Transition, MUUS Climate Partners, and Energy Impact Partners. He declares a significant financial interest in each company.

SUPPLEMENTAL INFORMATION

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REFERENCES

1. Smith, C., Hill, A.K., and Torrente-Murciano, L. (2020). Current and future role of haber–bosch ammonia in a carbon-free energy landscape. Energy Environ. Sci. *13*, 331–344. https://doi.org/10.1039/C9EE02873K.
2. International Energy Agency (2023). IEA Chemicals (International Energy Agency). https://www.iea.org/energy-system/industry/chemicals.
3. Department for Energy Security and Net Zero (2023). 2022 UK Greenhouse Gas Emissions, Provisional Figures (Department for Energy Security and Net Zero).
4. MacFarlane, D.R., Cherepanov, P.V., Choi, J., Suryanto, B.H.R., Hodgetts, R.Y., Bakker, J.M., Ferrero Vallana, F.M., and Simonov, A.N. (2020). A Roadmap to the Ammonia Economy. Joule *4*, 1186–1205. https://doi.org/10.1016/j.joule.2020.04.004.
5. International Maritime Organization (2023). 2023 IMO Strategy on Reduction of GHG Emissions from Ships (International Maritime Organization).
6. Valera-Medina, A., Amer-Hatem, F., Azad, A.K., Dedoussi, I.C., de Joannon, M., Fernandes, R.X., Glarborg, P., Hashemi, H., He, X., Mashruk, S., et al. (2021). Review on Ammonia as a Potential Fuel: From Synthesis to Economics. Energy Fuels *35*, 6964–7029. https://doi.org/10.1021/acs.energyfuels.0c03685.

7. Zhang, Y., Zanganeh, J., and Moghtaderi, B. (2025). Advancement in neutral and negative emissions production of ammonia from carbonaceous sources: a review. Int. J. Hydrogen Energy *109*, 40–62. https://doi.org/10.1016/j.ijhydene.2025.01.417.
8. Mayer, P., Ramirez, A., Pezzella, G., Winter, B., Sarathy, S.M., Gascon, J., and Bardow, A. (2023). Blue and green ammonia production: a techno-economic and life cycle assessment perspective. iScience *26*, 107389. https://doi.org/10.1016/j.isci.2023.107389.
9. Liu, X., Elgowainy, A., and Wang, M. (2020). Life cycle energy use and greenhouse gas emissions of ammonia production from renewable resources and industrial by-products. Green Chem. *22*, 5751–5761. https://doi.org/10.1039/D0GC02301A.
10. Wang, H., Zhou, P., Jeong, B., Mesbahi, A., Mujeeb-Ahmed, M.P., Jang, H., Giannakis, A., Sykaras, K., and Papadakis, A. (2025). Life cycle analysis of ammonia fuelled ship – case ship studies for marine vessels. J. Clean. Prod. *520*, 146105. https://doi.org/10.1016/j.jclepro.2025.146105.
11. Schreuder, W., Slootweg, J.C., and Van Der Zwaan, B. (2025). Techno-economic assessment of low-carbon ammonia as fuel for the maritime sector. Appl. Energy Combust. Sci. *22*, 100330. https://doi.org/10.1016/j.jaecs.2025.100330.
12. Dinh, Q.V., Pereira, P.H.T., Dinh, V.N., Nagle, A.J., and Leahy, P.G. (2024). Levelised cost of transmission comparison for green hydrogen and ammonia in new-build offshore energy infrastructure: pipelines, tankers, and HVDC. Int. J. Hydrogen Energy *62*, 684–698. https://doi.org/10.1016/j.ijhydene.2024.03.066.
13. Wong, A.Y.H., Selin, N.E., Eastham, S.D., Mounaïm-Rousselle, C., Zhang, Y., and Allroggen, F. (2024). Climate and air quality impact of using ammonia as an alternative shipping fuel. Environ. Res. Lett. *19*, 084002. https://doi.org/10.1088/1748-9326/ad5d07.
14. Wolfram, P., Kyle, P., Zhang, X., Gkantonas, S., and Smith, S. (2022). Using ammonia as a shipping fuel could disturb the nitrogen cycle. Nat. Energy *7*, 1112–1114. https://doi.org/10.1038/s41560-022-01124-4.
15. Li, J., Sun, Q., Yang, F., Wang, C., Feng, K., Xin, Y., Wang, W., and Li, D. (2025). Expanding the emissions trading system coverage can increase the cost competitiveness of low-carbon ammonia in China. Commun. Earth Environ. *6*, 231. https://doi.org/10.1038/s43247-025-02056-z.
16. D'Angelo, S.C., Cobo, S., Tulus, V., Nabera, A., Martín, A.J., Pérez-Ramírez, J., and Guillén-Gosálbez, G. (2021). Planetary Boundaries Analysis of Low-Carbon Ammonia Production Routes. ACS Sustainable Chem. Eng. *9*, 9740–9749. https://doi.org/10.1021/acssuschemeng.1c01915.
17. Li, J., Lin, J., Wang, J., Lu, X., Nielsen, C.P., McElroy, M.B., Song, Y., Song, J., Lyu, X., Yu, M., et al. (2025). Redesigning electrification of China's ammonia and methanol industry to balance decarbonization with power system security. Nat. Energy *10*, 762–773. https://doi.org/10.1038/s41560-025-01779-9.
18. Mingolla, S., Gabrielli, P., Manzotti, A., Robson, M.J., Rouwenhorst, K., Ciucci, F., Sansavini, G., Klemun, M.M., and Lu, Z. (2024). Effects of emissions caps on the costs and feasibility of low-carbon hydrogen in the European ammonia industry. Nat. Commun. *15*, 3753. https://doi.org/10.1038/s41467-024-48145-z.
19. Bose, A., Lazouski, N., Gala, M.L., Manthiram, K., and Mallapragada, D.S. (2022). Spatial Variation in Cost of Electricity-Driven Continuous Ammonia Production in the United States. ACS Sustainable Chem. Eng. *10*, 7862–7872. https://doi.org/10.1021/acssuschemeng.1c08032.
20. Liu, H., Zhou, S., Gu, W., Zhuang, W., Gao, M., Chan, C.C., and Zhang, X. (2025). Coordinated planning model for multi-regional ammonia industries leveraging hydrogen supply chain and power grid integration: a case study of shandong. Appl. Energy *377*, 124456. https://doi.org/10.1016/j.apenergy.2024.124456.
21. Wang, H., Daoutidis, P., and Zhang, Q. (2023). Ammonia-based green corridors for sustainable maritime transportation. Digit. Chem. Eng. *6*, 100082. https://doi.org/10.1016/j.dche.2022.100082.
22. Tu, R., Liu, C., Shao, Q., Liao, Q., Qiu, R., and Liang, Y. (2024). Pipeline sharing: optimal design of downstream green ammonia supply systems integrating with multi-product pipelines. Renew. Energy *223*, 120024. https://doi.org/10.1016/j.renene.2024.120024.
23. Xiang, P., Jiang, K., Wang, J., He, C., Jiang, W., and Chen, S. (2023). Zero-carbon ammonia technology pathway analysis by IPAC-technology model in China. J. Clean. Prod. *400*, 136652. https://doi.org/10.1016/j.jclepro.2023.136652.
24. Zhao, H., Kamp, L.M., and Lukszo, Z. (2021). Exploring supply chain design and expansion planning of China's green ammonia production with an optimization-based simulation approach. Int. J. Hydrogen Energy *46*, 32331–32349. https://doi.org/10.1016/j.ijhydene.2021.07.080.
25. RMI (2022). Transforming China's Chemicals Industry: Pathways and Outlook under the Carbon Neutrality Goal.
26. Li, Y., Lan, S., Ryberg, M., Pérez-Ramírez, J., and Wang, X. (2021). A quantitative roadmap for China towards carbon neutrality in 2060 using methanol and ammonia as energy carriers. iScience *24*, 102513. https://doi.org/10.1016/j.isci.2021.102513.
27. Xiang, P.-P., Wang, J.-C., Jiang, K.-J., He, C.-M., Jiang, W.-Y., Guo, L.-Q., Jiao, Y.-J., and Chen, S. (2025). Regional reallocation of zero-carbon ammonia production in China with carbon neutrality targets. Adv. Clim. Change Res. *16*, 199–212. https://doi.org/10.1016/j.accre.2025.01.001.
28. Atchison, J. (2025). Envision Energy: Successful Bunker Transfer Completed in Dalian (Ammonia Energy Association). https://ammoniaenergy.org/articles/envision-energy-successful-bunker-transfer-completed-in-dalian/.
29. Zhao, F., Fan, Y., Zhang, S., Eichhammer, W., Haendel, M., and Yu, S. (2022). Exploring pathways to deep de-carbonization and the associated environmental impact in China's ammonia industry. Environ. Res. Lett. *17*, 045029. https://doi.org/10.1088/1748-9326/ac614a.
30. IEA (2021). Ammonia Technology Roadmap (IEA).
31. Kang, L., Wu, W., Yu, H., and Su, F. (2022). Global container port network linkages and topology in 2021. Sensors *22*, 5889. https://doi.org/10.3390/s22155889.
32. Wang, Y., Chao, Q., Zhao, L., and Chang, R. (2022). Assessment of wind and photovoltaic power potential in China. Carb. Neutrality *1*, 15. https://doi.org/10.1007/s43979-022-00020-w.
33. Fan, J.-L., Xiang, X., Yao, Y., Li, K., Li, Z., Wei, S., Diao, Y., Ju, Z., Li, X., Li, X., et al. (2025). Dataset of CO2 geological storage potential and injection rate capacity in China based on fine grid technology. Sci. Data *12*, 640. https://doi.org/10.1038/s41597-025-04875-3.
34. Fasihi, M., and Breyer, C. (2024). Global production potential of green methanol based on variable renewable electricity. Energy Environ. Sci. *17*, 3503–3522. https://doi.org/10.1039/D3EE02951D.
35. Mingolla, S., and Rosa, L. (2025). Low-carbon ammonia production is essential for resilient and sustainable agriculture. Nat. Food *6*, 610–621. https://doi.org/10.1038/s43016-025-01125-y.
36. Esquivel-Elizondo, S., Walkowiak, B., Sartzetakis, S.S., and Buma, B. (2025). Climate Impact of Direct and Indirect N_2O Emissions from the Ammonia Marine Fuel Value Chain. Environ. Sci. Technol. *59*, 9037–9048. https://doi.org/10.1021/acs.est.4c13135.
37. Shafiee, R.T., and Schrag, D.P. (2024). Carbon abatement costs of green hydrogen across end-use sectors. Joule *8*, 3281–3289. https://doi.org/10.1016/j.joule.2024.09.003.
38. International Maritime Organization (2025). IMO Approves Net-Zero Regulations for Global Shipping (International Maritime Organization). https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-approves-netzero-regulations.aspx.
39. Sengupta, S. (2025). U.S. Pressures Countries to Delay Fees to Clean up Ship Pollution (The New York Times).
40. International Maritime Organization (2025). IMO Net-Zero Shipping Talks to Resume in 2026 (International Maritime Organization). https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-net-zero-shipping-talks-to-resume-in-2026.aspx.
41. Zahraee, S.M., and Shiwakoti, N. (2025). A review of sustainable hydrogen energy by 2050: supply chain, export markets, circular economy, social

dimensions, and future prospects: australia vs. worldwide. Sustain. Futures *10*, 101070. https://doi.org/10.1016/j.sftr.2025.101070.

42. Neumann, F., Hampp, J., and Brown, T. (2025). Green energy and steel imports reduce europe's net-zero infrastructure needs. Nat. Commun. *16*, 5302. https://doi.org/10.1038/s41467-025-60652-1.
43. IRENA (2025). Renewable Power Generation Costs in 2024 (IRENA).
44. Wood Mackenzie (2023). Should Gas Resource Holders Target LNG Exports or Blue Ammonia? (Wood Mackenzie).
45. Oxford Institute for Energy Studies (2023). The Outlook for China's Fossil Fuel Consumption under the Energy Transition and its Geopolitical Implications (Oxford Institute for Energy Studies).
46. Manigandan, S., Ryu, J.I., Praveen Kumar, T.R., and Elgendi, M. (2023). Hydrogen and ammonia as a primary fuel – A critical review of production technologies, diesel engine applications, and challenges. Fuel *352*, 129100. https://doi.org/10.1016/j.fuel.2023.129100.
47. Han, D., Liu, Y., and Huang, Z. (2022). The Use of Ammonia as a Fuel for Combustion Engines. In Engines and Fuels for Future Transport, G. Kalghatgi, A.K. Agarwal, F. Leach, and K. Senecal, eds. (Springer), pp. 233–256. https://doi.org/10.1007/978-981-16-8717-4_10.
48. Jafar, U., Nuhu, U., Khan, W.U., and Hossain, M.M. (2024). A review on green ammonia as a potential CO2 free fuel. Int. J. Hydrogen Energy *71*, 857–876. https://doi.org/10.1016/j.ijhydene.2024.05.128.
49. Xing, H., Stuart, C., Spence, S., and Chen, H. (2021). Alternative fuel options for low carbon maritime transportation: Pathways to 2050. J. Clean. Prod. *297*, 126651. https://doi.org/10.1016/j.jclepro.2021.126651.
50. Bertagni, M.B., Socolow, R.H., Martirez, J.M.P., Carter, E.A., Greig, C., Ju, Y., Lieuwen, T., Mueller, M.E., Sundaresan, S., Wang, R., et al. (2023). Minimizing the impacts of the ammonia economy on the nitrogen cycle and climate. Proc. Natl. Acad. Sci. USA *120*, e2311728120. https://doi.org/10.1073/pnas.2311728120.
51. Cai, B., Li, Q., and Zhang, X. (2021). China Status of CO2 Capture, Utilization and Storage (CCUS) 2021 — China's CCUS pathways (Chinese Academy of Environmental Planning, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, The Administrative Centre for China's Agenda 21).
52. Cai, B., Zhao, L., Zhang, Z., and Lu, X. (2023). China regional power grids carbon dioxide emission factors. In Ministry of Ecology and Environment of the People's Republic of China.
53. Zhuo, Z., Du, E., Zhang, N., Nielsen, C.P., Lu, X., Xiao, J., Wu, J., and Kang, C. (2022). Cost increase in the electricity supply to achieve carbon neutrality in China. Nat. Commun. *13*, 3172. https://doi.org/10.1038/s41467-022-30747-0.
54. RMI (2022). Transforming China's Chemicals Industry. https://rmi.org/insight/transforming-chinas-chemicals-industry/.
55. Zhengzhou Commodity Exchange (2019). Urea futures brochure. http://www.czce.com.cn/cn/ztlm/nsqhsszt/hygz/webinfo/2019/08/1564403229655432.htm.
56. National Bureau of Statistics of China (2021). China Statistical Yearbook (China Statistics Press).
57. Xiong, Y., Liu, W., Gao, P., Binqi, D., and Mingsheng, Z. (2022). Research on the hydrogen energy demand and carbon-reduction path in China's synthetic ammonia industry to achieve the "carbon peak" and "carbon neutrality" goals. Energy Storage Sci. Technol. *11*, 4048. https://doi.org/10.19799/j.cnki.2095-4239.2022.0364.
58. Cai, S., Zhao, X., Pittelkow, C.M., Fan, M., Zhang, X., and Yan, X. (2023). Optimal nitrogen rate strategy for sustainable rice production in China. Nature *615*, 73–79. https://doi.org/10.1038/s41586-022-05678-x.
59. Lim, J., Fernández, C.A., Lee, S.W., and Hatzell, M.C. (2021). Ammonia and Nitric Acid Demands for Fertilizer Use in 2050. ACS Energy Lett. *6*, 3676–3685. https://doi.org/10.1021/acsenergylett.1c01614.
60. Li, C., Liu, P., and Li, Z. (2022). A Long-Term Decarbonisation Modelling and Optimisation Approach for Transport Sector Planning Considering Modal Shift and Infrastructure Construction: A Case Study of China. Processes *10*, 1371. https://doi.org/10.3390/pr10071371.
61. Milani, D., Kiani, A., Haque, N., Giddey, S., and Feron, P. (2022). Green pathways for urea synthesis: A review from Australia's perspective. Sustain. Chem. Clim. Action *1*, 100008. https://doi.org/10.1016/j.scca.2022.100008.
62. Marlair, G., and Kordek, M.-A. (2005). Safety and security issues relating to low capacity storage of AN-based fertilizers. J. Hazard. Mater. *123*, 13–28. https://doi.org/10.1016/j.jhazmat.2005.03.028.
63. BfR (2020). Fillable articles made from melamine formaldehyde resin, such as coffee-to-go cups sold as 'bambooware', may leak harmful substances into hot foods (BfR-Stellungnahmen 2019). https://doi.org/10.17590/20200123-134155.